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Tandem: At-home Behavior Assessment Using Multimodal Signals from the Parent-Child Dyad

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The quality of parent-child interactions at an early age has been linked to children's social-emotional development, executive function, and risk for behavior problems. As such, parent-child interactions in naturalistic settings could present a unique opportunity to screen for at-risk behavior in young children, enabling timely and targeted interventions. In this work, we validate the feasibility of using structured at-home play sessions, completed via the Tandem smartphone app, to enable highly accurate and scalable behavioral assessments. We demonstrate that audio and physiological signals recorded during the play session can be used to capture key markers of parent-child interaction dynamics, which are more indicative of at-risk behavior compared to features from each individual alone. We propose novel audio-based dyadic interaction features that significantly outperform conventional speech features at predicting risk for behavior problems, achieving an F1 score of 0.87. Furthermore, we show that dyadic physiological synchrony features, extracted from privacy-preserving wearable sensor data, can classify at-risk behavior with an F1 score of 0.91. Tandem thus sets the stage for automated at-home behavior assessment tools for young children that balance screening accuracy with practical deployment considerations.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing**; **Human computer interaction (HCI)**; • **Applied computing → Health informatics**.

Additional Key Words and Phrases: mental health sensing, behavioral problems, parent-child interaction, audio, wearables

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1 INTRODUCTION

As humans, interacting with caregivers during early childhood is one of our most formative interpersonal experiences. Parenting practices and parent-child interactions at an early age are key drivers of children's verbal development and language acquisition [37, 78], academic readiness [67], executive function [80], and social-emotional development [48, 85]. As a result, there has been sustained and widespread interest in observing, quantifying, and improving early childhood parent-child interactions both within the child development [22, 32] and ubiquitous computing (UbiComp)/human-computer interaction (HCI) research communities [40, 42, 44, 45].

A large body of work in child development has especially focused on the association between parent-child interaction quality and behavior problems in young children. Communication patterns within the parent-child

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dyad, including maternal speech acts, child verbalizations, as well as dyadic interaction synchrony, have been shown to predict or moderate current and future behavior problems [16, 83, 95]. This association between the quality of young children's interactions with caregivers and their behavioral and emotional development presents a unique opportunity to screen children for behavior problems by observing parent-child dyadic interactions. Indeed, prior studies have investigated the feasibility of utilizing structured behavioral observations of parent-child interaction to screen children for behavior problems such as oppositional defiant disorder/conduct disorder [6] or attention deficit hyperactivity disorder (ADHD) [62]. At the same time, ubiquitous devices such as smartphones, smart speakers, and wearables open up new avenues for conducting these observations and assessments in naturalistic settings by recording parent-child interaction audio. Recent efforts in the UbiComp and HCI communities have laid the groundwork for automated observation tools that analyze parent-child speech (e.g., [44, 45, 84]). This has important implications for early-childhood mental health diagnosis, since identifying mood and behavior problems at a young age is known to be especially challenging. Specifically, before the age of five years, caregivers are often ill-equipped to differentiate early signs of mental health problems from normative misbehavior, such as temper tantrums, that are exhibited by all preschool-aged children[91]. Having access to ubiquitous tools that can leverage parent-child interactions in the wild to detect at-risk behavior can significantly alter the trajectory of diagnosis and treatment for young children and help improve long-term outcomes.

However, there is a significant gap in terms of in-the-wild screening technologies that leverage dyadic interactions to assess for behavior problems. Most existing work in the domain of technology and parent-child interactions has focused on sensing and improving the quality of interactions between the child and their caregivers, rather than examining the link between interaction quality and behavioral issues [44, 84]. Realizing the vision of interaction-based behavior assessments in the wild would therefore require a deeper investigation into (i) replicating the tasks and contexts that elicit parent/child behaviors predictive of underlying disorders outside clinical settings, and (ii) extracting relevant markers of dyadic interaction dynamics from in-the-wild observations to model behavioral issues. Additionally, using audio recordings to observe these interactions, particularly in home environments with children, could raise significant privacy concerns. This gives rise to a need for alternative sensing approaches that can capture dyadic interaction-based indicators of behavioral issues while preserving family privacy.

Our work, Tandem, addresses these gaps through the following contributions:

(1) *Translating and validating parent-child interaction-based clinical assessment paradigms in naturalistic settings:* We adapt a clinically validated analogue behavior observation, where parents complete a structured play session with their child, into a smartphone app-based assessment that can be completed in the home. The app records parent-child interaction audio that can be used to derive behavioral measures that are risk factors for behavior problems. We validate this assessment paradigm by conducting an at-home study with 68 participants, or 34 parent-child dyads, and demonstrate that it enables us to capture relevant information for in-the-wild behavior assessment.
(2) *Proposing novel dyadic interaction features for behavior assessment:* We record audio and wearable signals during the structured play session to extract features for behavior assessment. For audio signals, we introduce a novel temporal representation of parent-child conversational dynamics (see Section 3.2) that captures how parents and children respond to each other's speech acts. We compare these to frequency-based features and demonstrate that our proposed representations significantly improve classification performance from a baseline F1 score of 0.78 to 0.87. Our findings suggest that behavioral risk is manifested more strongly in dynamic conversational response patterns compared to singular speech behaviors from each participant.
(3) *Evaluating physiological signals as a privacy-preserving alternative to audio-based sensing:* We establish that physiological signals recorded during parent-child interaction can successfully capture autonomous

nervous system reactivity and dyadic physiological synchrony (see Section 3.3), which are known to reflect interaction dynamics and correlate with the risk for behavior problems in children (see Section 2.2). Further, we demonstrate that these physiological markers obtained from the child wearable alone can achieve comparable performance to audio-based features (F1 scores of 0.86 vs. 0.87). Our study is also the first to evaluate the feasibility of using physiological synchrony features from the parent-child dyad to screen for behavior problems in the wild – we show that, similar to the case of audio features, these markers of dyadic reactivity outperform individual features, achieving an F1 score of 0.91.

(4) *Validating abbreviated assessment protocols:* We show that data recorded during specific phases of the structured play session can achieve similar assessment performance as that from the entire session duration. We conduct a thorough evaluation of audio, physiological, and multimodal features from each phase to demonstrate fundamental differences in terms of how each modality captures the most informative features under different contexts. Specifically, audio features from the 5-minute clean-up phase and physiological features from the 10-minute child-led play achieved F1 scores in the 0.87 to 0.88 range, while multimodal features from the clean-up phase further outperform these with an F1 score of 0.92.

Tandem has important implications for future research and clinical practice in child mental health. First, our work reinforces the value of structured parent-child play interactions as an observational setting for at-home behavior assessments in young children. We demonstrate that behavioral assessment can be performed effectively using either interaction audio or physiological signals, with each modality capturing valuable markers of parent-child dynamics that classify at-risk behavior with an accuracy that is comparable or higher than commonly used screening instruments [53]. While our findings confirm the importance of dyadic measures in audio analysis via the use of novel interaction features, they also reveal the effectiveness of both dyadic and child-only physiological measures, offering practical flexibility for the implementation of assessment tools. Tandem thus provides a valuable foundation for future dyadic interaction sensing systems and opens new possibilities for widespread, accessible at-home screening tools that respect family privacy while providing valuable clinical insights. Future research in the UbiComp and HCI communities can build on these principles to develop and deploy scalable tools for at-home screening that balance assessment accuracy with practical deployment considerations.

2 RELATED WORK

2.1 Parent-Child Interaction Quality and Behavior Problems

The quality of children's interactions with their primary caregivers at a young age plays a critical role in the development of chronic self-regulation difficulties and subsequent behavior problems [24, 65]. Parent-child interaction synchrony has been shown to promote the development of self-regulation and adjustment strategies, while dysynchrony, emotional distress, or negative tone in these interactions increases children's risk for poor self-regulation and emotion socialization [23, 81, 83]. In recent years, research has shown that parent and child speech acts measured using the Dyadic Parent-Child Interaction Coding System (DPICS; [22]) are associated with internalizing and externalizing problems [6, 16, 33]. Zlomke et al. found significant differences in parent-child interaction behavior observed via DPICS codes across children with and without symptoms of autism spectrum disorder [95]. Keown found that paternal and maternal sensitivity, warmth, positive regard, and intrusive behavior were differentially associated with children's ADHD symptoms at preschool age and through middle childhood [49]. A dynamic systems analysis of parent-child interaction behavior found that behavioral rigidity, defined in terms of the durations of and transitions between positive, negative, and neutral parent and child behaviors, was associated with concurrent internalizing and externalizing behaviors as well as their progression over time [38]. Prior work has also found associations between parent mental health and children's current externalizing and internalizing problems [7] as well as future development of conduct problems [15], which were further mediated by the quality of the home environment and of parent-child interactions. In addition to the transactional processes

described above, developmental factors such as verbal deficits also indicate risk for child behavior problems. Lower linguistic maturity has been linked to higher risk of externalizing disorders [10, 41], while parent-reported communicative abilities are associated with both internalizing and externalizing behaviors [43].

However, despite the growing evidence of the link between parent-child interaction quality and the risk for diverse behavior problems, clinical diagnosis in early childhood still relies heavily on parent-reported diagnostic scales (e.g., the Behavior Assessment System for Children, BASC-3 [77]) or structured diagnostic interviews (e.g., the Preschool Age Psychiatric Assessment [18]). These assessments can be utilized to screen for a range of behavioral issues – for instance, the "Externalizing Problems" subscale of the BASC-3 measures an array of overt and disruptive behaviors, such as aggression, defiance, and attention/hyperactivity problems. Children who score on the extreme end of the BASC-3 may be at risk for DSM disorders such as ADHD, Oppositional Defiance Disorder, Conduct Disorder. On the other hand, the "Internalizing Problems" scale measures an array of mood and anxiety problems. Children who score at the extreme end of this scale are at risk for DSM disorders such as depression, dysthymia, and various anxiety disorders (separation anxiety disorder, generalized anxiety disorder, etc). In this work, we attempt to enable similar broadband screening for early childhood behavior problems using automatically extracted measures of parent-child interaction, thus reducing the reliance on subjectively reported measures for clinical diagnosis.

2.2 Physiological Correlates of Behavior Problems

The regulation of physiological arousal, especially in challenging task contexts, has been theorized to be a key component of children's self-regulation abilities that develop in early childhood, the lack of which can manifest as behavioral and emotional disorders [8, 17, 64]. Prior research has established links between parasympathetic nervous system activity (indexed by heart rate variability or HRV) and the regulation of emotions, attention, and behavior [9, 72]. Respiratory sinus arrhythmia (RSA), or the variability in heart rate within the respiratory frequencies, has also been associated with emotional reactivity [86]. Suppression of RSA in challenging contexts is further indicative of active and appropriate coping [73], while RSA dysregulation has been associated with ADHD, disruptive behavior disorders, and other behavior problems [17, 87, 89]. Similarly, researchers have also studied the relationship between behavior problems and sympathetic nervous system activity, often indexed by electrodermal activity (EDA). Morris et al. found that ADHD was associated with higher RSA and elevated EDA reactivity during an emotion suppression task [64]. Baker et al. found that low EDA levels during child compliance tasks predicted higher externalizing behavior [3]. In the context of parent-child interactions, Moffitt et al. found a link between more supportive parent reactions and scaffolding and higher baseline RSA and higher electrodermal reactivity in children on the autism spectrum [63]. Parental scaffolding during problem-solving tasks also moderated the relationship between EDA under-arousal and externalizing problems [3]. In light of these findings, our study recorded HR, HRV, RSA, and EDA signals from the parent and child in a dyadic interaction setting using wrist-worn wearables. We used these signals to obtain measures of parasympathetic and sympathetic autonomous nervous system reactivity that were in turn leveraged to predict risk for behavior problems.

In addition to individual physiological reactivity, recent work has shown that physiological synchrony, or the alignment between the physiological responses of the parent and child in a dyadic setting, has important implications for parent-child relationships and child behavior. Parent-child dyadic RSA synchrony has been shown to intensify the effect of parenting behaviors on youth behavior problems [66], and lower RSA synchrony during mild stress and free play has been linked to higher risk for internalizing and externalizing problems [59]. Lower parent-child EDA synchrony during free play has been associated with higher child autism spectrum disorder symptoms [4]. Our work further examines whether measures of heart rate, RSA, and EDA synchrony between parent and child during a structured play session can be utilized to screen children for behavior problems.

2.3 Parent-Child Interactions and Behavior Assessment in UbiComp Settings

There has been significant interest within the UbiComp and HCI community in observing and improving parent-child interactions in different settings [82]. TalkBetter [42] and TalkLIME [84] record parent-child interactions and derive metalinguistic features to provide parents feedback on their communication with children. SpecialTime [40] analyzes parent speech using the Dyadic Parent-Child Interaction Coding System (DPICS) to provide real-time feedback for families undergoing parent-child interaction therapy. Other systems facilitate parent-child communication during various activities of daily living [12, 44, 51, 93].

In terms of sensing approaches, most prior work reported above has focused on audio-based observations of parent-child communication. This has led to an investment in the development of datasets and models for analyzing parent-child speech [40, 45], although many applications still rely on commercially available audio processing techniques that are not tuned to child speech [92, 93]. A small number of prior studies have also used wearable sensing to monitor or facilitate parent-child interactions. For example, ParentGuardian used EDA sensing to monitor parental stress and deliver interventions via smartphone- and tablet-based apps [70]. FamilyScope used HRV and EDA during family interactions to provide feedback about emotional arousal and stress levels [54]. Chong et al. used wearable cameras to detect children's gaze and eye contact during interactions with adults [14].

While there has been considerable interest in behavioral assessments in young children using smartphones and wearables in home settings (e.g., [2, 46, 57]), there is limited work using parent-child interactions as a setting for naturalistic behavior assessment. The closest work is by Hernandez et al., who used EDA responses during a semi-structured play session to detect parent-reported ease of engagement with children [36]. Tandem extends this field of research by proposing flexible, dual-modal approaches using audio and/or physiological signals to enable behavioral screening during play-based parent-child interactions.

3 METHODS

3.1 In-home Parent-Child Interaction Study

3.1.1 Participants. To develop Tandem, we recruited 34 preschool-aged children (3- to 5-year-olds) and their parents (i.e., 68 total participants) to participate in an in-home parent-child interaction study. Parents were recruited through a developmental science research participant registry at the authors' institution and via community outreach and advertisements in the area. Parent-child dyads were eligible to participate as long as they spoke English as one of the languages at home and the child had not been diagnosed with any developmental delays or intellectual impairment. Children were not screened for, or selected based on, their history of behavior problems. 12 of the children were female and 22 were male, with a mean age of 4.3 years (SD: 1 year). The parent participating in the study was the biological mother (N=27), biological father (N=6), or adoptive mother (N=1) of the child. The child's race as reported by the parent was predominantly White (N=30), followed by Black or African American (N=3) and Asian (N=1), with one child identified as Hispanic or Latino. All study procedures were approved by the Institutional Review Board at the University of Massachusetts Amherst.

3.1.2 Study Application and Play Session. The parent and child together completed a 25-minute-long, at-home structured play session using the custom-built Tandem study app (see Figure 1). The app was built to be compatible with both Android and iOS devices using the Flutter framework [26]. Parents downloaded the app on their smartphones and reviewed a brief set of instructions before beginning the play session.

We chose to leverage play-based interactions as the basis of our at-home study task as they have long been established as an important setting for the development of parent-child relationships at a young age [29]. Several standardized protocols and manual observational measures have been designed to evaluate the quality of parent-child interactions in the context of play [22, 32], with some measures also linked to behavior problems [16].

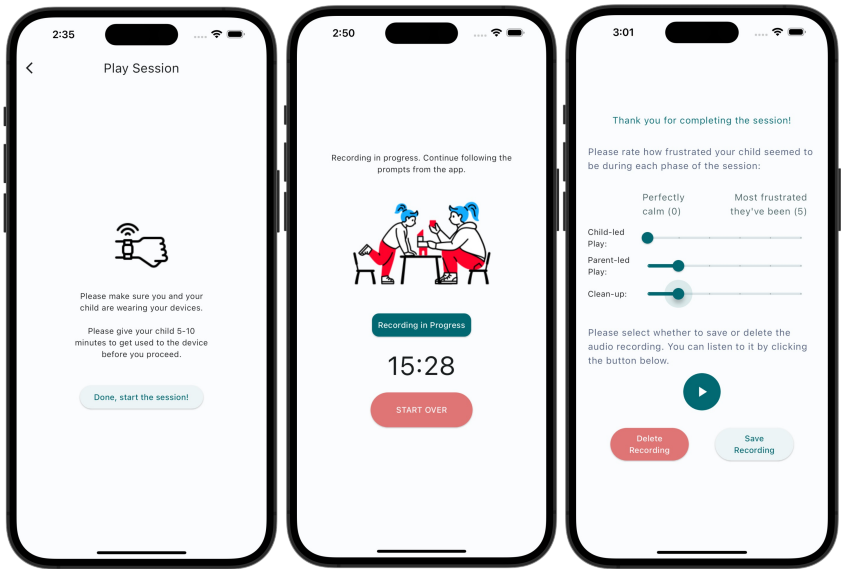

Fig. 1. Screenshots of the Tandem smartphone app, which was used to record parent-child interaction audio during the at-home play session.

The play session in our study was based on activities completed by families during Parent-Child Interaction Therapy (PCIT; [21]). PCIT is an evidence-based treatment for young children with behavior problems, where the therapist observes parent-child interactions in the clinic while they are engaged in a structured play session and provides parents with coaching on skills to manage their child's behavior. We adapted this session for our study, which consisted of three phases, each of which was initiated by an audio prompt from the app. The session began with ten minutes of child-led play (CLP), where the instructions asked parents to let their child choose any activity they want and play with them following their lead. The next ten minutes consisted of parent-led play (PLP), where the parent was asked to engage the child in playing according to rules set by the parent. Lastly, the clean-up (CU) phase asked parents to direct the child to put away all toys independently and without assistance. The CU phase lasted 5 minutes, after which the app informed the parent that the play session was complete.

The app recorded parent and child audio throughout the session. Parents were provided the opportunity to review their audio data before choosing whether to upload it to a secure server for inclusion in the study. They were also asked to rate their child's frustration levels during each phase (CLP, PLP, and CU) on a scale of 1-5.

3.1.3 Audio and Wearable Data. Parent and child audio was recorded via the study app throughout the play session in order to observe conversation behaviors similar to those that are recorded by PCIT therapists when observing parent-child interaction in real-time. Two-channel audio was recorded at a sampling rate of 44.1 kHz and a bitrate of 128 Kbps. Physiological signals (heart rate, respiration rate, electrodermal activity, and skin temperature) were recorded using an Empatica EmbracePlus wristband worn by the parent and child on their non-dominant wrist. Parents were asked to ensure that both themselves and their child put on their respective wristbands at least five minutes before starting the play session in order to record baseline physiological data and for children to get used to the device.

3.1.4 Parent-reported Behavior Scores. To obtain ground-truth data about children's risk for behavioral disorders, parents were asked to report their child's behavior on a set of clinically validated scales. For this work, we

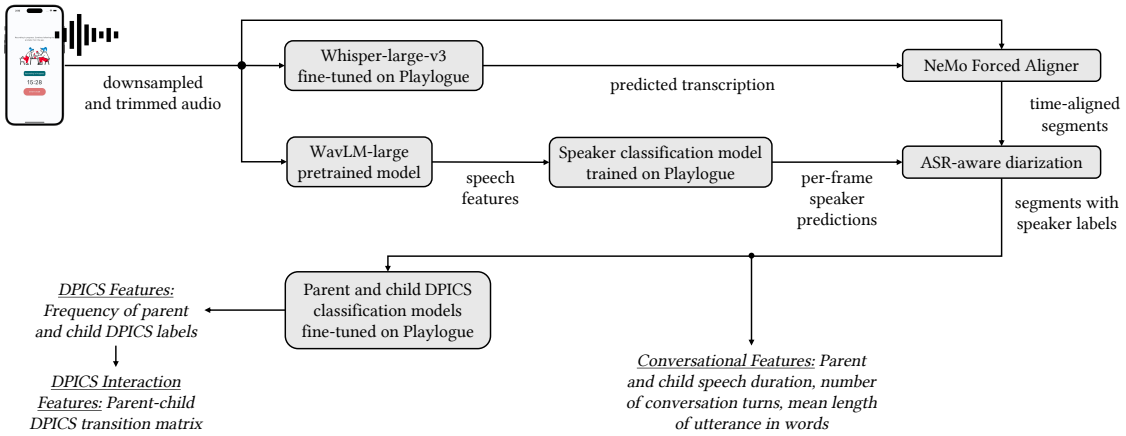

Fig. 2. Audio processing and feature extraction pipeline.

focus on parent-reported scores on the Behavior Assessment System for Children (BASC-3 [77]), which offers a comprehensive set of rating scales to measure children's behavior and emotions and compare these against age-based normative scores. The BASC-3 parent rating scales consist of 139 items describing possible behaviors, asking parents how frequently their child engaged in each behavior in the last several months. Participants rated each item on a 4-point Likert scale ("Never", "Sometimes", "Often", "Always").

The scores for each item were aggregated into 12 subscales, which were further condensed into raw scores on four standardized composite indices ("Externalizing Problems", "Internalizing Problems", "Behavioral Symptoms Index", and "Adaptive Skills"). The "Externalizing Problems" index provides a measure for children's aggression, defiance, disruptive behaviors, etc., and indicates risk for disorders such as ADHD and conduct disorder. The "Internalizing Problems" index is focused on providing measures of social stress, mood, and anxiety, quantifying risk for anxiety disorders, depression, etc. The "Behavioral Symptoms Index" provides an overview of a child's behavioral and emotional difficulties, including attention problems, conduct problems, and emotional dysregulation. Lastly, the "Adaptive Skills" index measures the child's ability to adapt to daily life situations, including social and academic skills. The raw scores on the four indices were converted to T scores using age-appropriate norms. A composite T score between 60 and 69 indicates at-risk behavior, while a T score of 70 or higher indicates clinically significant behavior issues on the former three indices. Similarly, a T score of 40 or lower on the "Adaptive Skills" index is considered at-risk, while that of 30 or lower is considered clinically significant [52, 77]. Note that these cut-offs are based on normed percentiles built from a large representative sample by the creators of the assessment instrument (e.g., a T score of 70 on the first index means the child scored at the 95th percentile for externalizing problems), and are the gold-standard in clinical assessment for determining whether a child is at risk for mental health disorders [47, 77].

For the analysis in this paper, we grouped all child participants with at-risk or clinically significant scores on at least one of the four composite indices into the "at-risk" category and others into the "not at risk" category. This resulted in 12 participants (35.3%) being categorized as "at-risk" and 20 (58.8%) as "not at risk", with 2 participants (5.9%) removed from the dataset due to missing responses on the BASC-3 assessment. All predictive analysis reported in Section 4 has therefore been conducted on 32 parent-child dyads.

3.2 Audio Processing and Feature Extraction

Figure 2 shows the audio processing pipeline comprising automatic speech recognition, time alignment, speaker diarization, and feature extraction. The following sections describe each of these steps in more detail.

3.2.1 Transcription and Speaker Diarization. The parent-child interaction audio was first automatically transcribed and diarized in preparation for feature extraction. We utilized child audio and speech benchmark models from the recent work on Playlogue [45] to process dyadic audio, since our participant age range is similar to that of this prior work. Audio recordings were downsampled from 44.1 kHz to 16 kHz. Audio from each play session is trimmed to form three separate clips, one for each phase of the session (CLP, PLP, or CU). The audio instructions from the app at the start of each phase were removed from the final clips, which contained only parent and child audio.

The OpenAI Whisper-large-v3 model [75], fine-tuned on the Playlogue dataset [45], was used to automatically transcribe parent-child audio recordings. Since Whisper does not provide accurate timestamps per segment (Whisper's word-level timestamps include silences between words/segments), the output transcript was forced-aligned to the audio using the NeMo Forced Aligner tool from the NVIDIA NeMo Framework [35].

To classify adult vs. child speech, the pre-trained WavLM-large model [13] was first used to extract per-frame speech embeddings. A multilayer perceptron model with two hidden layers was trained on the Playlogue dataset using WavLM features to predict speaker identities for each audio frame (parent vs. child). The model was then used to infer per-frame speaker identities for the parent-child interaction audio in the present study.

To further ensure accurate diarization, we implemented a novel ASR-aware diarization correction procedure that used the segment start and end timestamps from the aligned ASR transcripts. First, the per-frame speaker predictions from the diarization model were extracted for each frame within the aligned ASR segment. Since each segment represents a single sentence that is likely spoken by the same individual, the majority speaker label was assigned to the entire segment. The correction procedure was evaluated on the Playlogue test set, leading to a significant reduction in the overall diarization error rate (DER) from the 31.8% reported in [45] to 24.6%.

We then evaluated this modified pipeline on audio clips from three participants ($\sim$ 10%) in the present study that were manually transcribed and assigned ground-truth speaker labels. The pipeline achieved a DER of 30.1% and an ASR word error rate of 11.8% on these clips, demonstrating performance that is comparable or significantly better than that on the Playlogue test set as reported in [45]. The automated pipeline was also found to preserve most of the semantic content compared to ground truth, with a BERTScore [94] F1 of 0.97. Given the strong performance of the pipeline on the three validation clips, we used it to automatically transcribe and diarize segments from the rest of the data in preparation for downstream audio feature extraction.

3.2.2 Conversational Features. We first extracted "conversational features" from the transcribed parent-child audio that have previously been linked to child development and behavior to evaluate whether they would be predictive of behavior problems. The aligned start and end timestamps and speaker labels were used to compute the duration of parent and child speech over the entire session and within each phase. The diarization information was used to calculate the number of conversation turns, defined as a back-and-forth alternation between the parent and child (or vice versa) with no more than five seconds between them, which are associated with children's social-emotional development [30]. Finally, the mean length of utterance (MLU), a measure of children's gross language development [68], was calculated in terms of the mean number of words per utterance.

3.2.3 DPICS Features. Next, we extracted speech act features based on the Dyadic Parent-Child Interaction Coding System (DPICS; [22]). DPICS was first developed for use in PCIT to observe and record parent and child behaviors both during the pre-treatment baseline observation as well as to measure ongoing progress during therapy that focuses on improving parent-child interaction patterns. Each verbalization by the parent or child is assigned a distinct code, and the frequency of different DPICS codes during a session is used as a

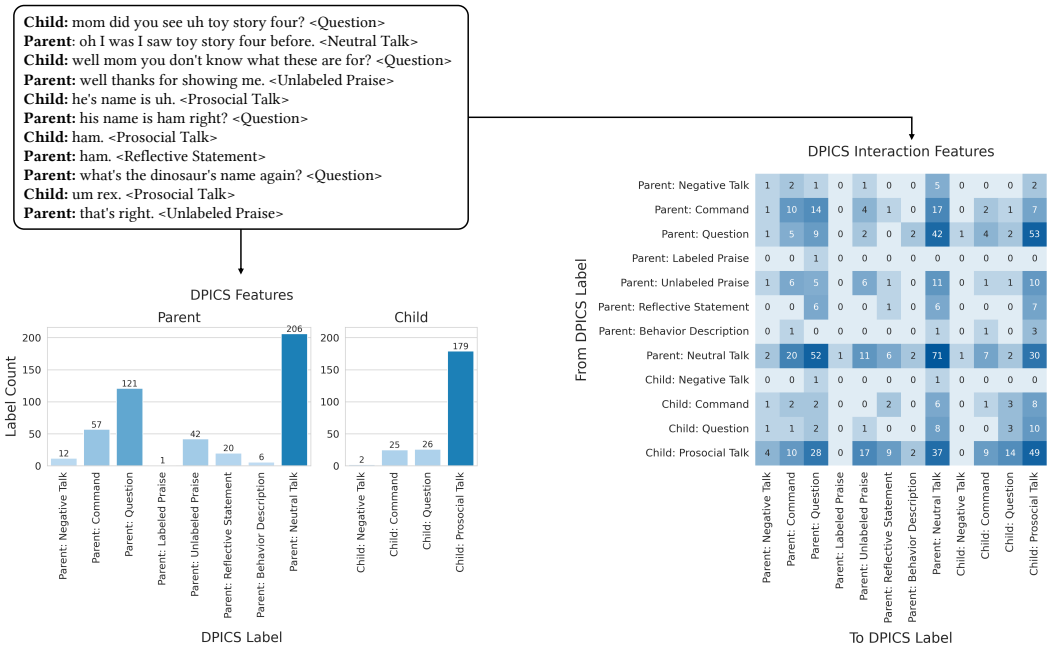

Fig. 3. Partial conversation transcript, DPICS features, and DPICS interaction features for a sample parent-child dyad.

measure of parent-child interaction quality. Following prior work [40, 45], we use the eight parent DPICS codes including "Negative Talk", "Command", "Labeled Praise", "Unlabeled Praise", "Question", "Reflective Statement", "Behavior Description", and "Neutral Talk" and the four child DPICS codes including "Negative Talk", "Command", "Question", and "Prosocial Talk" (see [22] for more details on DPICS). Parent DPICS codes have been widely used in parent-child interaction therapy to measure and improve parents' communication behavior with their young children [21], while both parent and child DPICS codes have been shown to be associated with behavior problems [6, 16].

We used sentence transformer models [76] trained on the Playlogue dataset with a context window of 5 consecutive utterances to extract DPICS parent and child codes for each transcribed segment from the play session. The ASR + diarization output was used as input to the parent and child DPICS models that were trained as described in the Playlogue publication [45]. Similar to the diarization and transcription pipelines, we validated the DPICS classification models by manually labeling three audio clips with DPICS codes and comparing these labels against the automated pipelines. The parent and child DPICS classification models were found to achieve an accuracy of 0.77 (8-class) and 0.91 (4-class), respectively, showing that these models could be reliably used for automated feature extraction.

The total number of occurrences of each DPICS label within the play session (as well as during each phase of the session) was extracted for use in predicting behavior problems. We henceforth refer to these as the "DPICS features" (see Figure 3).

3.2.4 DPICS Interaction Features. While the DPICS features described above provide a summary representation of parent and child speech acts during the play session, they fail to capture the temporal dynamics of the dyadic conversation. More specifically, DPICS counts do not capture any information about how the parent and child

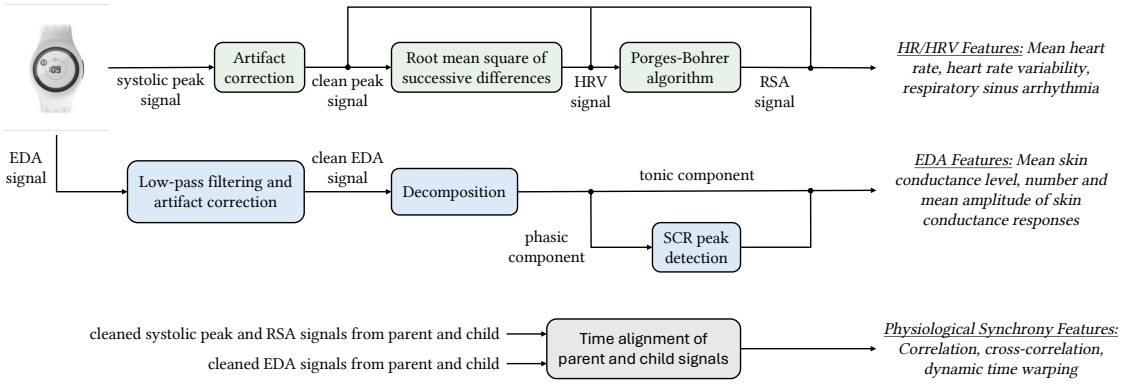

Fig. 4. Physiological signal processing and feature extraction pipeline.

respond to each other's speech over time. We therefore define a novel set of "DPICS interaction features" that represent the temporal and reactive variations in dyadic speech patterns.

The proposed DPICS interaction features are computed as a transition matrix T, where each element T_{ij} represents the number of instances wherein the DPICS label i was immediately followed by the DPICS label j within the play session (see Figure 3 for an example). This results in a 12x12 transition matrix encompassing all parent and child DPICS labels, which is then flattened into 144 individual features to be used in detecting behavior problems. Note that, since parent speech is classified into 8 DPICS codes and child speech into 4 DPICS codes, the transition matrix includes more parent-to-parent and child-to-child transitions (64 + 16 = 80 transitions) than parent-child or child-parent transitions (32 + 32 = 64 transitions). This means that the former intra-individual transitions are over-represented in the feature set compared to inter-individual transitions. However, the DPICS interaction features represent the *frequency* of each of these transitions – a higher number of intra-individual transitions indicates that the parent and child demonstrate less turn-taking behavior, which has been linked to developmental delays and behavior problems [19]. Similarly, a higher frequency of Child Negative Talk in response to Parent Command can indicate non-compliance and oppositional behavior. The DPICS interaction features capture these various measures in a single representation of dyadic behavior. Additionally, while these transition-based features still have some temporal locality by being restricted to sequences of only two immediate sentences, the overall frequency distribution of these transitions spanning a session provides valuable insight into both parent-child turn-taking behavior as well as both parties' immediate reactions to natural breakpoints in the conversation.

3.3 Wearable Data and Physiological Feature Extraction

3.3.1 Signal Preprocessing. Based on prior literature (see Section 2.2), heart rate and electrodermal activity (EDA) signals recorded via the EmbracePlus wristband were used to extract physiological features that could indicate risk for behavior problems. The systolic peak time series from the EmbracePlus was cleaned using the artifact correction algorithm proposed by Lipponen and Tarvainen [56]. EDA signals were cleaned using a low-pass filter with a cutoff of 1 Hz [74]. EDA artifacts were removed by discarding all 5-second segments where the signal changes by over 0.1 μS in less than 1 second [28]. Missing EDA segments that were less than 15 s long were interpolated using monotonic cubic interpolation, while longer chunks of missing data were discarded. The cleaned signals were used to extract individual physiological features from the parent and child as well as dyadic physiological synchrony features, as described in the following sections.

3.3.2 Individual Physiological Features. The cleaned systolic peak signal from each participant was used to extract heart rate (HR), heart rate variability (HRV), and respiratory sinus arrhythmia (RSA) features from the entire play session as well as for each phase of the session (where phase-level features were computed based on the signals recorded between the start and end times of each phase). Mean HR was extracted from peak differences after discarding consecutive peaks that were closer than 400 ms or farther than 1200 ms in order to exclude implausible heart periods based on normal human ranges[25, 61]. HRV was computed in terms of the root mean square of successive differences between peaks. Following recommendations from prior work [55], RSA was calculated using the Porges–Bohrer method [71] as the natural logarithm of the RSA signal variances over 30-second epochs. 4 out of 68 participants had inadequate length of filtered signals to extract phase-level HR, HR, and EDA features during at least one phase of the play session. However, session-level features were extracted successfully from all participants' data.

The cleaned EDA signal was decomposed into tonic and phasic components using the convex optimization approach proposed by Greco et al. [31]. The tonic component was used to extract the mean skin conductance level (SCL). The phasic component was used to extract skin conductance response (SCR) peaks using the algorithm proposed by Kim et al. [50]. The number of SCR peaks and the mean SCR peak amplitude were calculated for the whole session and for each phase. Due to signal quality issues, phase-level EDA features were missing for at least one of the three phases for 10 out of the 68 participants. Session-level EDA features were successfully extracted from all participants.

The above HR/HRV and EDA features together form the "individual physiological features" that were used to train models to detect behavior problems.

3.3.3 Physiological Synchrony Features. In addition to individual physiological features extracted from the systolic peak and electrodermal activity signals recorded by the parent and child's EmbracePlus devices, we computed measures of dyadic "physiological synchrony" that have been shown to indicate risk for behavior problems in prior work.

Following prior work, we computed the inter-beat interval (IBI) synchrony as the cross-correlation between parent and child IBI [1, 11, 34]. The individual IBI peak signals were filtered to remove intervals outside of the normal range of 0.4 to 1.2 seconds, aligned across participants, and interpolated and resampled to 1 Hz. We then fit an autoregressive integrated moving average model to eliminate autocorrelations and use the residuals to compute cross-correlation at a lag of 0. Additionally, to capture alignment between parent and child IBI at different time lags, we computed the distance between the normalized IBI series using Dynamic Time Warping (DTW) [27, 79]. To quantify dyadic RSA synchrony, we computed the Pearson correlation between time-aligned RSA signals from each participant, similar to other linear modeling approaches used in prior work on parent-child dyads [5, 66]. Finally, we estimated EDA synchrony by calculating the Pearson correlation between the time-aligned parent and child signals [4, 36]. We also computed the DTW distance between the normalized EDA signals from the parent and child [27, 36]. Physiological synchrony features were computed separately for each phase of the play session as well as for the overall session. 5 out of the 34 dyadic pairs had missing physiological synchrony features due to poor signal quality for either the child's or parent's wearable signals.

3.4 Modeling At-risk Behavior Using Audio and Physiological Signals

The audio features described in Section 3.2 (conversational, DPICS, and DPICS interaction features) and the wearable features described in Section 3.3 (individual physiology features and physiological synchrony features) were used to train and evaluate machine learning models to predict at-risk behavior in young children. We first evaluated whether each of these feature sets, with features extracted from the entire duration of the play session, was individually able to classify "at-risk" children from those "not at risk" based on their BASC-3 scores (see Section 3.1.4). When considering individual (i.e., not dyadic) features, we evaluated both child-only and

Table 1. Performance of Audio-based Classification Models (F1: F1 Score, Acc.: Accuray, Bal. Acc.: Balanced Accuracy).

	Child-only			Parent-only			Dyadic		
Audio Features	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.
Conversational features	-	-	-	-	-	-	0.70	0.81	0.77
DPICS features	0.78	0.84	0.83	0.63	0.78	0.73	0.78	0.84	0.83
DPICS interaction features	-	-	-	-	-	-	**0.87**	**0.91**	**0.89**

parent-only models as well as models using features from both the parent and the child audio or physiology. The results of this analysis are presented in Sections 4.1 and 4.2. Next, we evaluated multimodal models with audio and physiology features (Section 4.3).

We used the AutoGluon AutoML framework [20] to train and evaluate all our prediction models. We select this AutoML approach, which relies on ensembling and stacking multiple models in several layers, in order to avoid model selection and hyperparameter tuning that could lead to overfitting on our relatively high-dimensional, small sample-sized data. We also use AutoGluon's n-repeated k-fold bagging approach (with $n = 5$ and $k = 5$), which has been shown to be particularly effective for avoiding overfitting in smaller datasets [20] while also enabling us to compute out-of-fold cross-validation metrics indicative of unbiased model performance. More specifically, AutoGluon trains models on $k - 1$ folds of and performs inference on the held out fold, repeating the process on n random partitions of the data. These out-of-fold predictions are concatenated and compared to ground-truth values (rather than averaging across folds) to obtain unbiased estimates of model performance [20]. The train and validation folds contain disjoint parent-child dyads. We report model performance in terms of out-of-fold F1 score, accuracy, and balanced accuracy. To contextualize the model performances reported in Section 4, note that always predicting the majority class (i.e., "not at risk") would result in an F1 score of 0.77, accuracy of 0.63, and balanced accuracy of 0.50.

For the ensembling approach, we used all base models recommended by AutoGluon, excluding neural network-based models due to their tendency to overfit to small datasets (this was also empirically validated on our dataset). The set of base models thereby included LightGBM boosted trees, CatBoost boosted trees, random forests, extremely randomized trees, and k-nearest neighbors [20]. The final model selected by AutoGluon is a weighted ensemble of the base models. Sample weights were assigned based on the training data. Feature selection was performed via pruning with a prune ratio of 0.5 (i.e., pruning up to half of the original features at each step). Missing features were passed on to base models to be handled on a per-model basis.

4 RESULTS

4.1 Identifying At-risk Behavior Using Parent and Child Audio

Table 1 shows the performance of audio-based risk classification models trained using conversational features, DPICS features, and DPICS interaction features extracted from the entire length of the 25-minute dyadic play session. We see that basic conversational features such as speech duration and mean length of utterance can achieve a reasonable classification performance with an F1 score of 0.70. In terms of DPICS features, the 4 child DPICS labels alone achieve an F1 score of 0.78, whereas parent DPICS labels appear to be less predictive with an F1 score of only 0.63. Combining both parent and child DPICS features does not improve classification performance over only using child DPICS labels.

However, we observe that the novel DPICS interaction features proposed in Section 3.2.4 are more effective in capturing dyadic conversational markers that are predictive of behavior problems. The model trained using the

Table 2. Performance of Physiology-based Classification Models (F1: F1 Score, Acc.: Accuray, Bal. Acc.: Balanced Accuracy).

	Child-only			Parent-only			Dyadic		
Audio Features	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.
Individual physiology features	0.86	0.91	0.88	0.76	0.84	0.81	0.86	0.91	0.88
Physiological synchrony features	-	-	-	-	-	-	**0.91**	**0.94**	**0.92**

dyadic DPICS interaction features outperforms both conversational and DPICS count features, achieving an F1 score of 0.87 (accuracy = 0.91, balanced accuracy = 0.89) on out-of-fold predictions.

These findings reveal important insights about the mechanisms underlying parent-child interaction in at-risk dyads. The superior performance of DPICS interaction features over individual DPICS features provides empirical evidence supporting prior behavioral research that suggests that behavioral risk manifests not just in what parents and children say, but in how they respond to each other [39, 60]. Additionally, the relatively poor performance of parent-only DPICS features compared to child-only DPICS features may indicate that children's verbalization patterns are more directly indicative of their regulatory capabilities [69, 90] compared to parent speech acts.

4.2 Identifying At-risk Behavior Using Physiological Signals from the Parent and Child

We now investigate whether physiological features recorded during the dyadic play session (see Section 3.3) can be used to identify at-risk behavior. Table 2 shows the performance of classification models trained using individual physiology features from only the child, only the parent, or both the parent and child as well as the dyadic physiological synchrony features.

In evaluating the predictive performance of features from a single participant, we observe that physiology features from the HR/HRV and EDA signals from just the child can classify at-risk behavior with an F1 score of 0.86, while the same signals from the parent wearable achieve a lower F1 score of 0.76. Interestingly, physiological features from the child wearable alone can achieve classification performance comparable to that of the DPICS interaction features presented in Section 4.1. This finding holds important implications for child mental health researchers, demonstrating that it is possible to perform automated, at-home behavior assessments using a single off-the-shelf wearable device worn by the child as they complete a play session. It also eliminates the need to record sensitive audio data during the session, preserving the privacy of the child and their family.

Further evaluating physiology-based risk classification models, we see from Table 2 that the model trained on a combination of individual physiology features from both parent and child does not outperform the child physiology model, achieving the same F1 score, accuracy, and balanced accuracy of 0.86, 0.91, and 0.88, respectively. On the other hand, we find that dyadic physiological synchrony features significantly outperform individual features, achieving an F1 score of 0.91 (accuracy = 0.94, balanced accuracy = 0.92). This suggests that synchrony features such as correlation and lag-aware dynamic time warping distances are better at capturing co-regulation behaviors during parent-child interaction compared to individual features. Our results are also in line with prior work linking behavioral problems with parent-child physiological synchrony [4, 59].

While the physiological synchrony model outperforms the models trained on individual features, the synchrony features for this model necessitate the deployment of two wearable devices to record both parent and child data. Therefore, in resource-constrained settings, researchers and practitioners may want to instead rely on the child-only model that still achieves a reasonably high classification performance with an F1 score of 0.86.

Table 3. Performance of Multimodal Classification Models.

	F1 Score	Accuracy	Balanced Accuracy
DPICS interaction features only (baseline, from Table 1)	0.87	0.91	0.89
DPICS interaction features + child physiology features	0.80	0.88	0.83
DPICS interaction features + parent physiology features	0.87	0.91	0.89
DPICS interaction features + parent and child physiology features	**0.92**	**0.94**	**0.95**
DPICS interaction features + physiological synchrony features	0.86	0.91	0.88

4.3 Multimodal Audio+Physiology Models for Behavior Assessment

The analyses presented in the preceding sections demonstrate that at-home dyadic play sessions provide a valuable setting for automated behavior assessment, with both audio and physiological signals independently yielding highly informative features that are predictive of at-risk behavior. This enables child mental health researchers and practitioners to choose either modality for evaluation based on factors such as classification accuracy, ease of deployment, participant comfort, or privacy considerations.

While we observe that physiological features achieve classification performance that is comparable to or better than that of audio features, we hypothesize that the two observational modalities contain complementary information about the dyadic processes at play during parent-child interaction. Therefore, we evaluate whether it is possible to leverage both sets of features to train multimodal models for behavior assessment that are more performant than the audio- or physiology-based models presented previously. To incorporate audio information in these models, we use the DPICS interaction features that showed the best classification performance among audio-based classification models in Section 4.1. However, unlike dyadic audio features that are extracted from a single stream of audio recording on the phone, physiological signals are recorded via separate devices worn by the parent and child. Obtaining physiological features from both parent and child therefore necessitates the deployment of two devices, which can be expensive and challenging in real-world situations. In light of these considerations, we evaluate multimodal models with different physiology-based feature sets, viz., child physiology features only, parent physiology features only, individual physiology features from both participants, and dyadic physiological synchrony features.

Table 3 shows the performance of multimodal risk classification models using the different physiological feature combinations. We observe that adding individual physiological features from either the child or the parent does not improve classification performance over the audio-only baseline. In fact, the addition of child physiology features worsens predictive performance, even as child physiology features achieved an F1 score of 0.86 in isolation (see Table 2). This counterintuitive result likely reflects that the audio and physiological modalities capture overlapping aspects of the child's regulatory state through different mechanisms. More concretely, child physiological patterns and verbal interaction behaviors may both reflect emotional dysregulation, but may be prone to collinearity effects that make it difficult for the model to integrate them effectively.

However, we observe that adding individual physiology features from both participants results in more accurate predictions compared to the audio-only baseline. This suggests that having information on both parent and child physiological reactivity allows the model to learn complementary dyadic interaction patterns beyond what is available from verbal features alone. Interestingly, the physiological synchrony features do not have the same advantageous effect, suggesting that the model may be leveraging different physiological markers from the parent and child (e.g., EDA vs. HR) rather than the co-regulation of any one response.

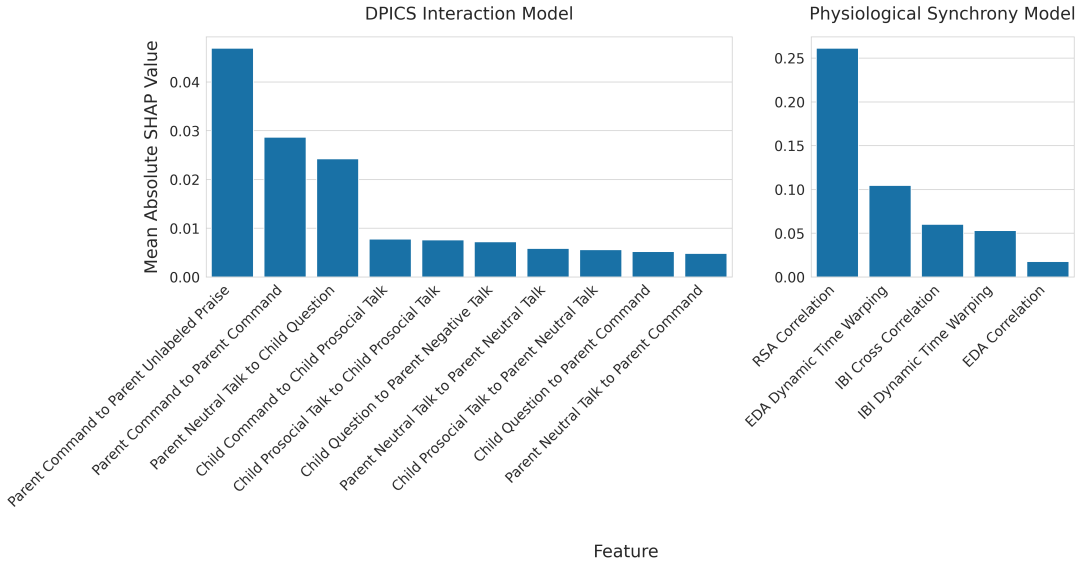

Fig. 5. Mean absolute SHAP values for the top-10 features in the audio-based classification model using DPICS interaction features as well as for all the features in the wearable-based classification model using physiological synchrony features.

Lastly, we note that the multimodal model with DPICS interaction + parent and child physiology features shows only a small improvement over the classification model using physiological synchrony features (F1 score of 0.92 vs. 0.91, see Table 2). Given our relatively small dataset and the out-of-fold validation approach used by AutoGluon, which does not return per-fold metrics, we do not have a way to evaluate whether this improvement is significant. Therefore, the utility of multimodal models over a physiology-only approach is still an open research question that we plan to address in future work.

4.4 Explaining Risk Predictions of Audio and Physiology Models

The results in the previous sections demonstrate the feasibility of utilizing audio and physiological signals from the dyadic play session to automate behavior assessments for children. Our analysis also identified which audio, physiological, and multimodal feature sets are likely to be most predictive of at-risk behavior by comparing different individual and dyadic feature sets.

We now probe the importance of individual features in our predictive models more closely by computing their SHAP values [58]. SHAP values, or SHapley Additive exPlanations, are a model-agnostic, game-theoretic approach to explain the predicted outcomes of machine learning models. A feature with a positive SHAP value increases the probability of the prediction, while a feature with a negative SHAP value contributes negatively to the prediction. The contributions of individual features are proportional to the magnitude of their SHAP values and are additive across the model. Note that, as a result of feature pruning, our final models are fit only on a subset of the original feature set. Only the features in this subset will contribute to the model's prediction and therefore have a non-zero SHAP value.

Figure 5 shows the mean absolute SHAP value for each feature in the best-performing audio and physiology model (for the sake of clarity, we limit the former plot to the top-10 features).. For the audio model trained with DPICS interaction features, we observe that the frequency of Parent Command to Parent Unlabeled Praise, Parent Command to Parent Command, and Parent Neutral Talk to Child Question transitions emerge as the

Table 4. Classification Performance Using Features from Individual Phases of the Parent-Child Interaction Task (F1: F1 Score, Acc.: Accuray, Bal. Acc.: Balanced Accuracy).

	Child-led Play			Parent-led Play			Clean-up		
	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.	F1	Acc.	Bal. Acc.
Audio									
DPICS interaction features	0.82	0.88	0.85	0.82	0.88	0.85	**0.87**	**0.91**	**0.89**
Physiology									
Individual physiology (IP) features	**0.88**	**0.91**	**0.91**	0.74	0.84	0.79	0.76	0.84	0.81
Physiological synchrony (PS) features	0.87	0.91	0.89	0.61	0.72	0.69	0.76	0.84	0.81
Multimodal									
DPICS interaction + IP features	0.82	0.88	0.85	0.82	0.88	0.85	0.80	0.88	0.83
DPICS interaction + PS features	0.82	0.88	0.85	0.78	0.84	0.83	**0.92**	**0.94**	**0.93**

most influential features, with mean SHAP values between 0.05 and 0.02. Parent Command also features most frequently in the top-10 features on par with Parent Neutral Talk, and may reflect intrusive parenting styles that disrupt children's autonomous play [88], a known risk factor for behavioral problems. While the novel DPICS interaction features have not been validated previously for their association with behavior problems, this finding is in line with prior work that found DPICS codes corresponding to negative talk and commands to be useful in screening for behavioral disorders [6].

For the physiological synchrony model, parent-child RSA correlation contributes most significantly to the final prediction, with a mean SHAP value of 0.26, followed by the EDA DTW distance, which has a mean SHAP value of 0.10. Both RSA and EDA synchrony have been linked to behavior problems and autism spectrum disorder [4, 66], showing that our model predictions are grounded in known physiological synchrony-based risk factors.

4.5 Behavior Assessment Using Signals from Individual Phases of the Play Session

Our analyses thus far have focused on behavior assessment using dyadic audio, physiology, or multimodal features extracted from signals from the entire 25-minute play session. However, given the structured nature of the play session and the varying demands on parents and children during each phase, we hypothesize that the interaction patterns exhibited by the parent-child dyad are likely to be unique to each phase. Additionally, in terms of behavior assessment, we hypothesize that the interaction dynamics from each of the three phases are not equally predictive of at-risk behavior.

To test this hypothesis, we trained and evaluated AutoML models using features computed independently from the signals recorded during each individual phase of the play session (i.e., child-led play or CLP, parent-led play or PLP, and clean-up or CU). Informed by our previous results, we consider the following feature sets for this analysis: (i) DPICS interaction features, (ii) dyadic physiology features as well as physiological synchrony features, and (iii) DPICS interaction + dyadic physiology features and DPICS interaction + physiological synchrony features.

Table 4 shows the out-of-fold classification performance of audio, physiology, and multimodal models trained using the respective features from singular phases of the dyadic play session. We observe that CU audio models outperform other stages, achieving an F1 score of 0.87 and an accuracy of 0.91. Interestingly, this is on par with the performance of audio models trained using features from the entire play session (see Table 1). This suggests that it may be enough for mental health practitioners to design just a 5-minute, structured clean-up task that provides enough conversational context for at-home, audio-based behavior assessments. However, given that the CU phase in our study follows 20 minutes of other structured activity that may influence dyadic interaction

behavior during this phase, further research is required to validate the feasibility of using it as a standalone assessment activity.

In a similar vein, we observe that the model trained using individual physiology features from both parent and child during the CLP phase achieves an F1 score of 0.88 , outperforming models trained using individual physiology features from other phases. The model trained on physiological synchrony features from the CLP phase also demonstrates comparable performance with an F1 score of 0.87. Given that the CLP phase is scheduled during the start of the play session and is therefore unlikely to be influenced by the rest of the session, this demonstrates the feasibility of using only a 10-minute assessment task to predict at-risk behavior using parent and child wearable signals with a reasonably high accuracy. However, the performance of phase-level models using physiological features is still slightly lower than that of the task-level model from Table 2, (F1 score = 0.91). This suggests that the full length of the task may provide more valuable signals for behavior assessment due to both the longer time context as well as the opportunity to observe physiological synchrony between parent and child within more diverse settings.

Evaluating phase-level multimodal models, we find that the DPICS interaction + physiological synchrony features obtained from the CU phase outperform all others with an F1 score of 0.92 (accuracy = 0.94, balanced accuracy = 0.93). Similar to the audio-only scenario, this multimodal model achieves a classification performance that is comparable to the multimodal model using data from the entire 25-minute play session (see Table 3).

Overall, our results suggest that the parent-child interaction dynamics at play during each phase of the session, and the underlying behavioral risk factors, are probably best captured using different sensing modalities. Audio and multimodal sensing appear to glean the most discriminative information to classify at-risk children from the CU phase, with multimodal models outperforming the audio-only scenario. On the other hand, physiological features observed during CLP are most revealing of interaction-based risk factors.These phase-specific patterns have important implications for assessment design: rather than collecting data across all interaction contexts, future screening tools can strategically match sensing modalities to the behavioral constructs most likely to manifest in each phase, potentially enabling shorter, more targeted assessments that preserve clinical utility while reducing burden on the family.

4.6 Evaluating the Feasibility of Predicting Multi-dimensional Behavior Symptoms

As described in Section 1, Tandem primarily focuses on behavior assessment by identifying high-level at-risk behavior that can then be probed further by child mental health practitioners. In this regard, Tandem acts as a broadband screening mechanism that does not aim to evaluate or score children on specific behavioral dimensions, instead classifying above vs. below threshold scores on a range of combined scales as discussed in Section 3.1.4.

However, we recognize that researchers and clinicians may be interested in gaining more specific insight into children's behavior along different evaluation axes using at-home screening, since this can enable targeted intervention strategies to address particular issues. To this end, we conduct follow-up analyses investigating the feasibility of leveraging audio and physiological features from the dyadic play session to predict children's T scores on each of the four BASC-3 standardized composite indices (i.e., "Externalizing Problems", "Internalizing Problems", "Behavioral Symptoms Index", and "Adaptive Skills"). We used the AutoGluon framework to train regression models using audio (DPICS interaction features) and physiology (physiological synchrony features) feature sets that were found to be the best-performing ones at the classification tasks using the entire duration of the play session. The models are trained and evaluated using a similar setup as was used for the classification models in previous sections, with the performance reported in terms of mean absolute error, normalized root-mean-square error, and Pearson correlation between actual and predicted scores. To help contextualize model performance, we also report the metrics for a baseline model predicting a T score of 50 (population mean) for every participant.

Table 5. Performance of Regression Models for Predicting Scores on Each Standardized Composite Index in BASC-3.

	Mean Absolute Error	Normalized RMSE	Pearson Correlation
Baseline - Mean Prediction			
Externalizing Problems	4.91	0.24	-
Internalizing Problems	7.75	0.30	-
Behavioral Symptoms	5.62	0.27	-
Adaptive Skills	5.06	0.18	-
DPICS Interaction Features			
Externalizing Problems	4.40	0.12	$r = 0.45, p < 0.01$
Internalizing Problems	5.80	0.12	$r = 0.56, p < 0.001$
Behavioral Symptoms	4.41	0.10	$r = 0.63, p < 0.001$
Adaptive Skills	4.00	0.13	$r = 0.24, p = 0.19$
Physiological Synchrony Features			
Externalizing Problems	2.68	0.12	$r = 0.90, p < 0.001$
Internalizing Problems	4.69	0.20	$r = 0.66, p < 0.001$
Behavioral Symptoms	4.19	0.22	$r = 0.60, p < 0.001$
Adaptive Skills	3.55	0.15	$r = 0.59, p < 0.001$

Table 5 shows the performance of the audio- and physiology-based regression models predicting each BASC-3 index. We observe that, on average, DPICS interaction features can predict scores on all BASC-3 indices to within 4.00 to 5.80 points of the actual score, add achieve mean absolute errors well below the baseline on all indices except "Externalizing Problems". Except for those on the "Adaptive Skills" index, the predicted scores also exhibit significant positive correlations with actual scores. On the other hand, regression models trained using physiological synchrony features demonstrate mean absolute errors between the range of 2.68 to 4.69, which is slightly lower than those of the audio-based models for all subscales, with a considerable improvement for the "Externalizing Problems" index. The predicted scores from the physiology-based models are also significantly correlated with actual scores on all indices, with the strongest positive correlation of 0.90 for scores on the "Externalizing Problems" index positive correlationsbetween 0.59 and 0.66 on the other three indices. Overall, our results indicate that physiological synchrony features outperform audio features when predicting exact scores on a range of behavior dimensions. However, the predictive performance differs for each index, and further research should examine the feasibility of using different feature sets to predict scores on various outcomes of interest as well as the clinical utility of such tools at a given level of regression accuracy.

5 DISCUSSION

5.1 Parent-Child Interactions as a Naturalistic yet Controlled Setting for Behavior Assessments

As described previously, a large body of work in the child development and the UbiComp research communities has hitherto focused on parent-child interactions as an important, but standalone, outcome to measure and improve [40, 42, 44, 93]. Our work extends this perspective by demonstrating that structured parent-child interactions can serve a dual purpose, enabling highly accurate behavior assessment based on dyadic interaction dynamics.

We selected play-based parent-child interactions as the basis for Tandem as they provide an ideal balance between controlled assessment (such as in clinical settings) and ecological validity. The structured nature of the interaction allows us to establish a more comparable setup across participants when studying child mental health in naturalistic environments. Furthermore, the play session is based on clinically validated observational settings used in parent-child interaction therapy [22], and prior research links interaction behaviors captured in this setting to various developmental outcomes [16].

Our findings demonstrate that interaction dynamics during this structured play session are indeed valuable for predicting behavior problems. The clean-up phase proved particularly revealing of at-risk behavior when using audio-based DPICS interaction features, achieving performance matching that of the entire session with an F1 score of 0.87. This suggests that specific task structures can elicit diagnostically valuable behavioral patterns, which aligns with clinical practices in parent-child interaction therapy [21] while making them accessible outside formal settings. The findings open new directions for exploring shorter, more targeted assessment protocols that could be easily integrated into home routines while maintaining clinical validity.

From a practical standpoint, parent-child interactions have several characteristics that make them an ideal setting for behavioral assessments. First, leveraging day-to-day interactions in the home makes behavioral screenings much more accessible and scalable compared to clinical visits. The ability to collect relevant information about interaction dynamics using commercial, off-the-shelf devices such as smartphones and wearables further facilitates convenient and scalable assessments. Using play-based interactions, in particular, also makes the screening process feel more natural to the child [46]. Lastly, as shown in Section 2, there is strong evidence linking parent-child interaction patterns to behavior problems. Utilizing simple, interpretable features that are known to be associated with behavioral outcomes allows researchers to develop explainable predictive models as well as design targeted interventions to then improve behavioral outcomes. These findings highlight the potential for leveraging parent-child interaction behavior as a promising approach for in-the-wild assessments of child behavior problems.

5.2 Physiological Signals as an Alternative to Audio-based Interaction Analysis

While audio sensing has dominated parent-child interaction research in UbiComp settings [40, 42, 84], our findings show that physiological signals provide more informative features and stronger predictive performance for behavior assessment in young children (F1 score of 0.87 for audio vs. 0.91 for physiology). Our physiological sensing approach successfully detected dyadic patterns associated with at-risk behavior through measures of autonomic nervous system activity, aligning with prior research showing associations between physiological reactivity and behavior problems [8].

This represents a significant paradigm shift for measuring parent-child interaction dynamics and their association with behavior problems. Physiological data can be collected continuously and unobtrusively through commercially available smartwatches and wearable devices. It offers a privacy-preserving alternative that eliminates the need to record potentially sensitive audio content. Wearable-based assessment is also a compelling alternative in multilingual households or when parent-child interactions may be interrupted by other family members, where audio processing presents additional challenges.

The strong performance of child-only physiological features (F1 score of 0.86) is also particularly significant, as it has profound implications for scaling assessment approaches. This finding suggests that it is possible to achieve strong predictive performance using a single wearable device worn by the child, rather than requiring the more challenging and resource-intensive deployment of two separate wearables for both participants.

Our multimodal analysis further demonstrated that combining physiological and audio features may yield marginally stronger predictive performance (F1 score of 0.92), although the added utility of multimodal signals needs to be further researched. The ability to achieve high accuracy with physiological signals alone, however,

creates new opportunities for continuous monitoring that could transform early intervention approaches by identifying behavior concerns before they become entrenched.

5.3 Ethical Considerations for In-the-Wild Deployment

The deployment of sensor-based behavioral assessment tools in home environments raises important ethical questions that need to be carefully considered.

First, recording audio data during parent-child interactions in home environments presents substantial privacy concerns. The intimate nature of parent-child play sessions conducted at home means that audio data inevitably includes private family moments that the families may not have intended to share. However, our findings represent a significant advance in the privacy-aware deployment of such assessment tools. We find that physiological signals alone achieve comparable (F1 score of 0.86 using child wearable features) or better (F1 score of 0.91 using synchrony features) predictive performance compared to audio features (F1 score of 0.87), making them a privacy-preserving alternative that enables families to participate in behavioral screening while maintaining greater control over their personal information.

Second, even though Tandem achieves a high predictive performance, sensor-based behavioral assessment cannot solely be considered an objective indicator of behavioral risk independent of context. Parent-child interaction patterns vary significantly across cultural and socioeconomic contexts, which must be accounted for when diagnosing or treating children for behavior problems. Our binary classification approach, while useful for screening purposes, does not account for the complex, multidimensional nature of these interactions. In addition, our study population was based in the United States and was predominantly white, which may limit the generalizability of our machine learning models to other geographical and social contexts.

In light of these considerations, the responsible deployment of machine learning-based tools like Tandem requires several safeguards. Clinical oversight must remain central to any screening process, with sensor-based assessments serving to augment rather than replace professional judgment. We expect the deployment of such sensor-based assessments to be done with a clinical provider in the loop to ensure that families are provided the appropriate context and support needed after using the tool.

5.4 Opportunities for Future Work

While Tandem is a significant proof of concept for the ability to use parent-child interaction dynamics for behavior assessment in the wild, there are opportunities for future research to both validate and improve upon the findings presented in this work. First, while we extensively validate our feature extraction pipelines and machine learning models, we do not optimize for the highest performing models via approaches such as hyperparameter tuning and model selection. We instead take a conservative approach by using ensemble models and repeated k-fold cross-validation to avoid overfitting given our relatively small sample size. Future work can explore more complex modeling approaches to further improve classification performance.

Second, there is also an opportunity for future research to dig deeper into the DPICS interaction and physiological synchrony features that we validate here for behavior assessment. Our investigation of phase-level models demonstrates that structured parent-child interaction sessions of shorter durations (5 or 10 minutes) could provide enough context to derive useful dyadic features. Future work could explore the limits on the minimum length of interaction required to obtain informative audio and/or physiological features. Similarly, researchers can examine various ways to effectively combine dyadic features with the individual-level features used to compute them, which would require approaches to handle multicollinearity and feature interactions. The performance of multimodal models could also be analyzed more closely and in larger populations to determine whether audio and physiological signals provide complementary information for behavior assessment.

6 CONCLUSION

This paper presents Tandem – a suite of models for automated behavior assessment in young children using audio and/or physiological signals recorded during in-home, play-based, parent-child interactions. We conducted a study with 68 participants (34 parent-child dyads), who completed a structured play session with child-led, parent-led, and clean-up phases. We developed pipelines to extract meaningful markers of dyadic interaction patterns and synchrony from smartphone-based audio and wearable-based physiological signals from both parent and child recorded during the play session. Our analysis revealed that either audio or physiological features can independently achieve high accuracy in detecting at-risk behavior, with audio-based DPICS interaction features achieving an F1 score of 0.87 and physiological synchrony features achieving an F1 score of 0.91. This establishes the feasibility of utilizing privacy-preserving physiological features for behavior assessment while maintaining high accuracy, and provides researchers and clinicians the flexibility to choose between modalities based on practical considerations. Our work has important implications for the child mental health and UbiComp research communities, demonstrating the value of parent-child interaction dynamics during structured play interactions for scalable behavior assessment.

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References

[1] Nigela Ahemaitijiang, Haining Ren, Hui Wang, and Zhuo Rachel Han. 2021. Longitudinal association between emotion-related parenting behavior and child aggression: The moderating role of parent-child physiological synchrony. *Aggressive behavior* 47, 3 (2021), 267–275.

[2] Riku Arakawa, Karan Ahuja, Kristie Mak, Gwendolyn Thompson, Sam Shaaban, Oliver Lindhiem, and Mayank Goel. 2023. LemurDx: Using Unconstrained Passive Sensing for an Objective Measurement of Hyperactivity in Children with no Parent Input. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 2 (June 2023), 1–23. doi:10.1145/3596244

[3] Jason K Baker, Rachel M Fenning, Stephen A Erath, Brian R Baucom, Jacquelyn Moffitt, and Mariann A Howland. 2018. Sympathetic under-arousal and externalizing behavior problems in children with autism spectrum disorder. *Journal of Abnormal Child Psychology* 46 (2018), 895–906.

[4] Jason K Baker, Rachel M Fenning, Mariann A Howland, Brian R Baucom, Jacquelyn Moffitt, and Stephen A Erath. 2015. Brief report: A pilot study of parent–child biobehavioral synchrony in autism spectrum disorder. *Journal of autism and developmental disorders* 45 (2015), 4140–4146.

[5] Samantha L Birk, Lindsey Stewart, and Thomas M Olino. 2022. Parent–child synchrony after early childhood: A systematic review. *Clinical Child and Family Psychology Review* 25, 3 (2022), 529–551.

[6] Åse Bjørseth, Cheryl McNeil, and Lars Wichstrøm. 2015. Screening for behavioral disorders with the Dyadic Parent-Child Interaction Coding System: Sensitivity, specificity, and core discriminative components. *Child & Family Behavior Therapy* 37, 1 (2015), 20–37.

[7] Kaitlin Bountress and Laurie Chassin. 2015. Risk for behavior problems in children of parents with substance use disorders. *American Journal of Orthopsychiatry* 85, 3 (2015), 275.

[8] Susan D Calkins. 2009. Regulatory competence and early disruptive behavior problems: The role of physiological regulation. *Biopsychosocial Regulatory Processes in the Development of Childhood Behavioral Problems* (2009).

[9] Susan D Calkins, Paulo A Graziano, and Susan P Keane. 2007. Cardiac vagal regulation differentiates among children at risk for behavior problems. *Biological psychology* 74, 2 (2007), 144–153.

[10] Susan B Campbell, Susan Spieker, Margaret Burchinal, Michele D Poe, and NICHD Early Child Care Research Network. 2006. Trajectories of aggression from toddlerhood to age 9 predict academic and social functioning through age 12. *Journal of Child Psychology and Psychiatry* 47, 8 (2006), 791–800.

[11] Yasemin Zehra Capraz, Kerstin Konrad, and Vanessa Reindl. 2023. Concurrent and lagged physiological synchrony during mother–child interaction and their relationship to positive affect in 8-to 10-year-old children. *Scientific Reports* 13, 1 (2023), 17744.

[12] Meng-Ying Chan, Yi-Hsuan Lin, Long-Fei Lin, Ting-Wei Lin, Wei-Che Hsu, Chia-yu Chang, Rui Liu, Ko-Yu Chang, Min-hua Lin, and Jane Yung-jen Hsu. 2017. WAKEY: assisting parent-child communication for better morning routines. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 2287–2299.

[13] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing* 16, 6 (2022), 1505–1518.

[14] Eunji Chong, Katha Chanda, Zhefan Ye, Audrey Southerland, Nataniel Ruiz, Rebecca M Jones, Agata Rozga, and James M Rehg. 2017. Detecting gaze towards eyes in natural social interactions and its use in child assessment. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 3 (2017), 1–20.

[15] Andrea M Chronis, Benjamin B Lahey, William E Pelham Jr, Stephanie Hall Williams, Barbara L Baumann, Heidi Kipp, Heather A Jones, and Paul J Rathouz. 2007. Maternal depression and early positive parenting predict future conduct problems in young children with attention-deficit/hyperactivity disorder. *Developmental psychology* 43, 1 (2007), 70.

[16] Allison M Cotter and Elizabeth Brestan-Knight. 2020. Convergence of parent report and child behavior using the Dyadic Parent-Child Interaction Coding System (DPICS). *Journal of Child and Family Studies* 29, 11 (2020), 3287–3301.

[17] Lourdes P Dale, Emily A O'Hara, Rachel Schein, Lorinn Inserra, Julie Keen, Marci Flores, and Stephen W Porges. 2011. Measures of infant behavioral and physiological state regulation predict 54-month behavior problems. *Infant Mental Health Journal* 32, 4 (2011), 473–486.

[18] HL Egger, BARBARA H Ascher, and A Angold. 1999. Preschool age psychiatric assessment (PAPA). *Durham (North Carolina): Duke University Medical Center* (1999).

[19] Jennifer Harrison Elder and Jill Jordan Goodman. 1996. Social turn-taking of children with neuropsychiatric impairments and their parents. *Issues in Comprehensive Pediatric Nursing* 19, 4 (1996), 249–261.

[20] Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. 2020. Autogluon-tabular: Robust and accurate automl for structured data. *arXiv preprint arXiv:2003.06505* (2020).

[21] SM Eyberg and B Funderburk. 2011. Parent-child interaction therapy protocol. *Gainesville, FL: PCIT International* (2011).

[22] Sheila M Eyberg, MM Nelson, Maura Duke, and Stephen R Boggs. 2004. Manual for the dyadic parent-child interaction coding system third edition. *Unpublished Rating Manual* (2004).

[23] Ruth Feldman, Charles W Greenbaum, and Nurit Yirmiya. 1999. Mother–infant affect synchrony as an antecedent of the emergence of self-control. *Developmental psychology* 35, 1 (1999), 223.

[24] Tiffany Field. 1994. The effects of mother's physical and emotional unavailability on emotion regulation. *Monographs of the society for research in child development* (1994), 208–227.

[25] Susannah Fleming, Matthew Thompson, Richard Stevens, Carl Heneghan, Annette Plüddemann, Ian Maconochie, Lionel Tarassenko, and David Mant. 2011. Normal ranges of heart rate and respiratory rate in children from birth to 18 years of age: a systematic review of observational studies. *The Lancet* 377, 9770 (2011), 1011–1018.

[26] Flutter. [n. d.]. Flutter - Build apps for any screen. https://flutter.dev/. Accessed: 2024-08-15.

[27] Shkurta Gashi, Elena Di Lascio, and Silvia Santini. 2019. Using unobtrusive wearable sensors to measure the physiological synchrony between presenters and audience members. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 1 (2019), 1–19.

[28] Shkurta Gashi, Elena Di Lascio, Bianca Stancu, Vedant Das Swain, Varun Mishra, Martin Gjoreski, and Silvia Santini. 2020. Detection of artifacts in ambulatory electrodermal activity data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 2 (2020), 1–31.

[29] Kenneth R Ginsburg, Committee on Psychosocial Aspects of Child, Family Health, et al. 2007. The importance of play in promoting healthy child development and maintaining strong parent-child bonds. *Pediatrics* 119, 1 (2007), 182–191.

[30] Esteban Gómez Muzzio. 2022. Preliminary Findings: Early Turn Talking Is Associated with Later Social-Emotional Development. *Volta Review* 122, 1 (2022).

[31] Alberto Greco, Gaetano Valenza, Antonio Lanata, Enzo Pasquale Scilingo, and Luca Citi. 2015. cvxEDA: A convex optimization approach to electrodermal activity processing. *IEEE transactions on biomedical engineering* 63, 4 (2015), 797–804.

[32] Rebecca Grzadzinski, Themba Carr, Costanza Colombi, Kelly McGuire, Sarah Dufek, Andrew Pickles, and Catherine Lord. 2016. Measuring changes in social communication behaviors: preliminary development of the Brief Observation of Social Communication Change (BOSCC). *Journal of autism and developmental disorders* 46 (2016), 2464–2479.

[33] Katherine Anne Halas. 2019. Maternal Borderline Personality Disorder and Negative Talk, and Young Children's Behavioral Symptoms.

[34] Zhuo Rachel Han, Mengyu M Gao, Julia Yan, Xiaoyi Hu, Wen Zhou, and Xiaomei Li. 2019. Correlates of parent-child physiological synchrony and emotional parenting: Differential associations in varying interactive contexts. *Journal of Child and Family Studies* 28 (2019), 1116–1123.

[35] E Harper, S Majumdar, O Kuchaiev, L Jason, Y Zhang, E Bakhturina, V Noroozi, S Subramanian, K Nithin, H Jocelyn, et al. [n. d.]. NeMo: A toolkit for conversational AI and large language models. https://nvidia.github.io/NeMo/. Accessed: 2024-08-15.

[36] Javier Hernandez, Ivan Riobo, Agata Rozga, Gregory D Abowd, and Rosalind W Picard. 2014. Using electrodermal activity to recognize ease of engagement in children during social interactions. In *Proceedings of the 2014 ACM international joint conference on pervasive and ubiquitous computing*. 307–317.

[37] Erika Hoff. 2003. The specificity of environmental influence: Socioeconomic status affects early vocabulary development via maternal speech. *Child development* 74, 5 (2003), 1368–1378.

[38] Tom Hollenstein, Isabela Granic, Mike Stoolmiller, and James Snyder. 2004. Rigidity in parent—child interactions and the development of externalizing and internalizing behavior in early childhood. *Journal of abnormal child psychology* 32 (2004), 595–607.

[39] Caroline P Hoyniak, Laura E Quiñones-Camacho, M Catalina Camacho, Jenna H Chin, Elizabeth M Williams, Lauren S Wakschlag, and Susan B Perlman. 2021. Adversity is linked with decreased parent-child behavioral and neural synchrony. *Developmental Cognitive Neuroscience* 48 (2021), 100937.

[40] Bernd Huber, Richard F. Davis, Allison Cotter, Emily Junkin, Mindy Yard, Stuart Shieber, Elizabeth Brestan-Knight, and Krzysztof Z. Gajos. 2019. SpecialTime: Automatically Detecting Dialogue Acts from Speech to Support Parent-Child Interaction Therapy. In *Proceedings of the 13th EAI International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth'19)*. ACM, 139–148. doi:10.1145/3329189.3329203

[41] Claire Hughes, Adele White, Joanna Sharpen, and Judy Dunn. 2000. Antisocial, angry, and unsympathetic:"Hard-to-manage" preschoolers' peer problems and possible cognitive influences. *The Journal of Child Psychology and Psychiatry and Allied Disciplines* 41, 2 (2000), 169–179.

[42] Inseok Hwang, Chungkuk Yoo, Chanyou Hwang, Dongsun Yim, Youngki Lee, Chulhong Min, John Kim, and Junehwa Song. 2014. TalkBetter: family-driven mobile intervention care for children with language delay. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing (CSCW'14)*. ACM, 1283–1296. doi:10.1145/2531602.2531668

[43] Rianne Jansen, Jarymke Maljaars, Anouk Verhappen, Inge Zink, Jean Steyaert, and Ilse Noens. 2020. Problem behavior in young children referred with language difficulties: Relations to language and intentional communication. *Autism & Developmental Language Impairments* 5 (2020), 2396941519900076.

[44] Eunkyung Jo, Hyeonseok Bang, Myeonghan Ryu, Eun Jee Sung, Sungmook Leem, and Hwajung Hong. 2020. MAMAS: Supporting Parent–Child Mealtime Interactions Using Automated Tracking and Speech Recognition. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW1 (May 2020), 1–32. doi:10.1145/3392876

[45] Manasa Kalanadhabhatta, Mohammad Mehdi Rastikerdar, Tauhidur Rahman, Adam S Grabell, and Deepak Ganesan. 2024. Playlogue: Dataset and benchmarks for analyzing adult-child conversations during play. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 4 (2024), 1–34.

[46] Manasa Kalanadhabhatta, Adrelys Mateo Santana, Zhongyang Zhang, Deepak Ganesan, Adam S Grabell, and Tauhidur Rahman. 2022. EarlyScreen: Multi-scale Instance Fusion for Predicting Neural Activation and Psychopathology in Preschool Children. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2 (July 2022), 1–39. doi:10.1145/3534583

[47] Randy W Kamphaus and CR Reynolds. 2015. Behavior assessment system for children. *The Clinical Assessment of Children and Adolescents: A Practitioner's Handbook* (2015).

[48] Ioannis Katsantonis and Ros McLellan. 2024. The role of parent–child interactions in the association between mental health and prosocial behavior: Evidence from early childhood to late adolescence. *International Journal of Behavioral Development* 48, 1 (2024), 59–70.

[49] Louise J Keown. 2012. Predictors of boys' ADHD symptoms from early to middle childhood: The role of father–child and mother–child interactions. *Journal of abnormal child psychology* 40 (2012), 569–581.

[50] Kyung Hwan Kim, Seok Won Bang, and Sang Ryong Kim. 2004. Emotion recognition system using short-term monitoring of physiological signals. *Medical and biological engineering and computing* 42 (2004), 419–427.

[51] Wonjung Kim, Seungchul Lee, Seonghoon Kim, Sungbin Jo, Chungkuk Yoo, Inseok Hwang, Seungwoo Kang, and Junehwa Song. 2020. Dyadic mirror: Everyday second-person live-view for empathetic reflection upon parent-child interaction. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 3 (2020), 1–29.

[52] Evelyn R Klein, Cesar E Ruiz, Kylee Morales, and Paige Stanley. 2019. Variations in parent and teacher ratings of internalizing, externalizing, adaptive skills, and behavioral symptoms in children with selective mutism. *International journal of environmental research and public health* 16, 21 (2019), 4070.

[53] John V Lavigne, Kathryn Mendelsohn Meyers, and Marissa Feldman. 2016. Systematic review: Classification accuracy of behavioral screening measures for use in integrated primary care settings. *Journal of pediatric psychology* 41, 10 (2016), 1091–1109.

[54] Hyunsoo Lee, Yugyeong Jung, Youwon Shin, Hyesoo Park, Woohyeok Choi, and Uichin Lee. 2024. FamilyScope: Visualizing Affective Aspects of Family Social Interactions using Passive Sensor Data. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (2024), 1–27.

[55] Gregory F Lewis, Senta A Furman, Martha F McCool, and Stephen W Porges. 2012. Statistical strategies to quantify respiratory sinus arrhythmia: are commonly used metrics equivalent? *Biological psychology* 89, 2 (2012), 349–364.

[56] Jukka A Lipponen and Mika P Tarvainen. 2019. A robust algorithm for heart rate variability time series artefact correction using novel beat classification. *Journal of medical engineering & technology* 43, 3 (2019), 173–181.

[57] Bryn C Loftness, Julia Halvorson-Phelan, Aisling O'Leary, Carter Bradshaw, Shania Prytherch, Isabel Berman, John Torous, William L Copeland, Nick Cheney, Ryan S McGinnis, et al. 2023. The ChAMP App: A Scalable mHealth Technology for Detecting Digital Phenotypes of Early Childhood Mental Health. *IEEE Journal of Biomedical and Health Informatics* (2023).

[58] Scott M Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems 30*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (Eds.). Curran Associates, Inc., 4765–4774. http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf

[59] Erika Lunkenheimer, Catherine M Hamby, Frances M Lobo, Pamela M Cole, and Sheryl L Olson. 2020. The role of dynamic, dyadic parent–child processes in parental socialization of emotion. *Developmental psychology* 56, 3 (2020), 566.

[60] Erika S Lunkenheimer, Sheryl L Olson, Tom Hollenstein, Arnold J Sameroff, and Charlotte Winter. 2011. Dyadic flexibility and positive affect in parent–child coregulation and the development of child behavior problems. *Development and psychopathology* 23, 2 (2011), 577–591.

[61] Dominique Makowski, Tam Pham, Zen J. Lau, Jan C. Brammer, François Lespinasse, Hung Pham, Christopher Schölzel, and S. H. Annabel Chen. 2021. NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods* 53, 4 (feb 2021), 1689–1696. doi:10.3758/s13428-020-01516-y

[62] Franziska Minder, Agnieszka Zuberer, Daniel Brandeis, and Renate Drechsler. 2018. A review of the clinical utility of systematic behavioral observations in attention deficit hyperactivity disorder (ADHD). *Child Psychiatry & Human Development* 49 (2018), 572–606.

[63] Jacquelyn M Moffitt, Jason K Baker, Rachel M Fenning, Stephen A Erath, Daniel S Messinger, Sasha M Zeedyk, Sarah A Paez, and Sydney Seel. 2021. Parental socialization of emotion and psychophysiological arousal patterns in children with autism spectrum disorder. *Research on child and adolescent psychopathology* 49 (2021), 401–412.

[64] Stephanie SJ Morris, Erica D Musser, Rachel B Tenenbaum, Anthony R Ward, Jessica Martinez, Joseph S Raiker, Erika K Coles, and Cameron Riopelle. 2020. Emotion regulation via the autonomic nervous system in children with attention-deficit/hyperactivity disorder (ADHD): replication and extension. *Journal of abnormal child psychology* 48 (2020), 361–373.

[65] Sheryl L Olson, Arnold J Sameroff, Erika S Lunkenheimer, and David C Kerr. 2009. Self-regulatory processes in the development of disruptive behavior problems: The preschool-to-school transition. In *Biopsychosocial regulatory processes in the development of childhood behavioral problems*. Cambridge University Press, 144–185.

[66] Assaf Oshri, Sihong Liu, Cynthia M Suveg, Margaret O'Brien Caughy, and Landry Goodgame Huffman. 2023. Biological sensitivity to context as a dyadic construct: An investigation of child–parent RSA synchrony among low-SES youth. *Development and psychopathology* 35, 1 (2023), 95–108.

[67] Faith Lamb Parker, Alison Y Boak, Kenneth W Griffin, Carol Ripple, and Lenore Peay. 1999. Parent-child relationship, home learning environment, and school readiness. *School Psychology Review* 28, 3 (1999), 413–425.

[68] Matthew D Parker and Kent Brorson. 2005. A comparative study between mean length of utterance in morphemes (MLUm) and mean length of utterance in words (MLUw). *First language* 25, 3 (2005), 365–376.

[69] Isaac T Petersen, John E Bates, Brian M D'Onofrio, Claire A Coyne, Jennifer E Lansford, Kenneth A Dodge, Gregory S Pettit, and Carol A Van Hulle. 2013. Language ability predicts the development of behavior problems in children. *Journal of abnormal psychology* 122, 2 (2013), 542.

[70] Laura Pina, Kael Rowan, Asta Roseway, Paul Johns, Gillian R Hayes, and Mary Czerwinski. 2014. In situ cues for ADHD parenting strategies using mobile technology. In *Proceedings of the 8th international conference on pervasive computing technologies for healthcare*. 17–24.

[71] Stephen W Porges and Robert E Bohrer. 1990. The analysis of periodic processes in psychophysiological research. (1990).

[72] Stephen W Porges, Jane A Doussard-Roosevelt, and Ajit K Maiti. 1994. Vagal tone and the physiological regulation of emotion. *Monographs of the society for research in child development* (1994), 167–186.

[73] Stephen W Porges, Jane A Doussard-Roosevelt, A Lourdes Portales, and Stanley I Greenspan. 1996. Infant regulation of the vagal "brake" predicts child behavior problems: A psychobiological model of social behavior. *Developmental psychobiology* 29, 8 (1996), 697–712.

[74] Hugo F Posada-Quintero and Ki H Chon. 2020. Innovations in electrodermal activity data collection and signal processing: A systematic review. *Sensors* 20, 2 (2020), 479.

[75] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*. PMLR, 28492–28518.

[76] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084* (2019).

[77] Cecil R Reynolds, Randy W Kamphaus, and Kimberly J Vannest. 2015. *BASC3: Behavior assessment system for children*. PscyhCorp Bloomington.

[78] Clare R Rogers, Karissa L Nulty, Mariana Aparicio Betancourt, and Laura S DeThorne. 2015. Causal effects on child language development: A review of studies in communication sciences and disorders. *Journal of communication disorders* 57 (2015), 3–15.

[79] Stan Salvador and Philip Chan. 2007. Toward accurate dynamic time warping in linear time and space. *Intelligent data analysis* 11, 5 (2007), 561–580.

[80] Valarie M Schroeder and Michelle L Kelley. 2010. Family environment and parent-child relationships as related to executive functioning in children. *Early child development and care* 180, 10 (2010), 1285–1298.

[81] Daniel S Shaw, Richard Q Bell, and Miles Gilliom. 2000. A truly early starter model of antisocial behavior revisited. *Clinical child and family psychology review* 3 (2000), 155–172.

[82] Ji Youn Shin, Minjin Rheu, Jina Huh-Yoo, and Wei Peng. 2021. Designing Technologies to Support Parent-Child Relationships: A Review of Current Findings and Suggestions for Future Directions. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (Oct. 2021), 1–31. doi:10.1145/3479585

[83] Cynthia L Smith, Susan D Calkins, Susan P Keane, Arthur D Anastopoulos, and Terri L Shelton. 2004. Predicting stability and change in toddler behavior problems: contributions of maternal behavior and child gender. *Developmental psychology* 40, 1 (2004), 29.

[84] Seokwoo Song, Seungho Kim, John Kim, Wonjeong Park, and Dongsun Yim. 2016. TalkLIME: mobile system intervention to improve parent-child interaction for children with language delay. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '16)*. ACM, 304–315. doi:10.1145/2971648.2971650

[85] Mai Stafford, Diana L Kuh, Catharine R Gale, Gita Mishra, and Marcus Richards. 2016. Parent–child relationships and offspring's positive mental wellbeing from adolescence to early older age. *The journal of positive psychology* 11, 3 (2016), 326–337.

[86] Cynthia A Stifter and Nathan A Fox. 1990. Infant reactivity: Physiological correlates of newborn and 5-month temperament. *Developmental psychology* 26, 4 (1990), 582.

[87] Rachel B Tenenbaum, Erica D Musser, Joseph S Raiker, Erika K Coles, Elizabeth M Gnagy, and William E Pelham. 2018. Specificity of reward sensitivity and parasympathetic-based regulation among children with attention-deficit/hyperactivity and disruptive behavior disorders. *Journal of abnormal child psychology* 46 (2018), 965–977.

[88] Germaine YQ Tng and Hwajin Yang. 2024. Indirect effects of intrusive parenting on externalizing behaviors in socioeconomically disadvantaged children: A parallel mediation analysis. *Cognitive Development* 72 (2024), 101516.

[89] Stella Tsotsi, Jessica L Borelli, Mumtaz Backer, Noraini Veragoo, Nurshuhadah Abdulla, Kok Hian Tan, Yap Seng Chong, Helen Chen, Michael J Meaney, Birit Broekman, et al. 2023. Preschoolers' emotion reactivity and regulation: Links with maternal psychological distress and child behavior problems. *Development and psychopathology* 35, 3 (2023), 1079–1091.

[90] Claire Vallotton and Catherine Ayoub. 2011. Use your words: The role of language in the development of toddlers' self-regulation. *Early childhood research quarterly* 26, 2 (2011), 169–181.

[91] Lauren S Wakschlag, Seung W Choi, Alice S Carter, Heide Hullsiek, James Burns, Kimberly McCarthy, Ellen Leibenluft, and Margaret J Briggs-Gowan. 2012. Defining the developmental parameters of temper loss in early childhood: implications for developmental psychopathology. *Journal of Child Psychology and Psychiatry* 53, 11 (2012), 1099–1108.

[92] Wenjie Xu, Jiayi Ma, Jiayu Yao, Weijia Lin, Chao Zhang, Xuanhe Xia, Nan Zhuang, Shitong Weng, Xiaoqian Xie, Shuyue Feng, et al. 2023. MathKingdom: Teaching Children Mathematical Language Through Speaking at Home via a Voice-Guided Game. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.

[93] Ying Xu, Kunlei He, Valery Vigil, Santiago Ojeda-Ramirez, Xuechen Liu, Julian Levine, Kelsyann Cervera, and Mark Warschauer. 2023. "Rosita Reads With My Family": Developing A Bilingual Conversational Agent to Support Parent-Child Shared Reading. In *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*. 160–172.

[94] Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations*. https://openreview.net/forum?id=SkeHuCVFDr

[95] Kimberly R Zlomke, Sarah Bauman, and Garet S Edwards. 2019. An exploratory study of the utility of the dyadic parent-child interaction coding system for children with Autism Spectrum Disorder. *Journal of Developmental and Physical Disabilities* 31 (2019), 501–518.